8086 / Microprocessor Math Revision Sheet

1. Real Mode Segment:Offset Math

Formula: Physical Address = Segment × 10H + Offset

Example 1: CS=1200H, IP=3400H → 1200H×10H=12000H; 12000H+3400H=15400H

Example 2: DS=2500H, SI=1A20H → 25000H+1A20H=26A20H

2. Reverse Segment:Offset Calculation

Formula: Offset = Physical Address – (Segment × 10H)

Example 1: Physical=12345H, Segment=1200H → Offset=0345H

Example 2: Physical=56789H, Segment=5600H → Offset=0789H

3. Protected Mode Selector / Descriptor Math

Formula: Descriptor Address = GDT Base + (Index × 8)

Example 1: Selector=0028H, GDT Base=1000H → Index=5 → Address=1028H

Example 2: Selector=0030H, GDT Base=2000H → Index=6 → Address=2030H

4. Short Jump Address Calculation

Formula: Target = Next Instruction Address + Displacement

Example 1: 0100: JZ +05H → Next=0102H → Target=0107H

Example 2: 0200: JNZ +0AH → Next=0202H → Target=020CH

5. LOOP Instruction Count

LOOP decrements CX and jumps while CX ≠ 0.

Example 1: CX=5 → loop executes 5 times.

Example 2: CX=3 → loop executes 3 times.

6. DJNZ Instruction Count (8051)

DJNZ decrements register and jumps while value ≠ 0.

Example 1: R0=4 → executes 4 times.

Example 2: R2=7 → executes 7 times.

7. Time / Delay Calculation

Formula: Delay = Number of Cycles × Clock Period

Example 1: 12 MHz clock → T=1/12MHz=83.33 ns. 100 cycles ≈ 8.33 µs

Example 2: 1 MHz clock → T=1 µs. 500 cycles = 500 µs